

Write a program to implement multithreaded program using pthreads Thread

A thread is a single sequence stream within a process. Because threads have some of the properties of processes, they are sometimes called *lightweight processes*.

Differences between process and thread?

Threads are not independent from each other unlike processes. As a result, threads shares with other threads their code section, data section and OS resources like open files and signals. But, like processes, a thread has its own program counter (PC), a register set, and a stack space.

PROGRAM:

```
#include <pthread.h>
#include <stdio.h>
#include <stdlib.h>
#include <sys/types.h>
#include <unistd.h>

// Let us create a global variable to change it in threads
int g = 0;

// The function to be executed by all threads
void* myThreadFun(void* vargp)
{
    // Store the value argument passed to this thread
    int myid = getpid();

    // Let us create a static variable to observe its
    // changes
    static int s = 0;

    // Change static and global variables
    ++s;
    ++g;

    // Print the argument, static and global variables
```

```
        printf("Thread ID: %d, Static: %d, Global: %d\n", myid,
               ++s, ++g);
    }

int main()
{
    int i;
    pthread_t tid;

    // Let us create three threads
    for (i = 0; i < 3; i++)
        pthread_create(&tid, NULL, myThreadFun, NULL);

    pthread_exit(NULL);
    return 0;
}
```

Output: